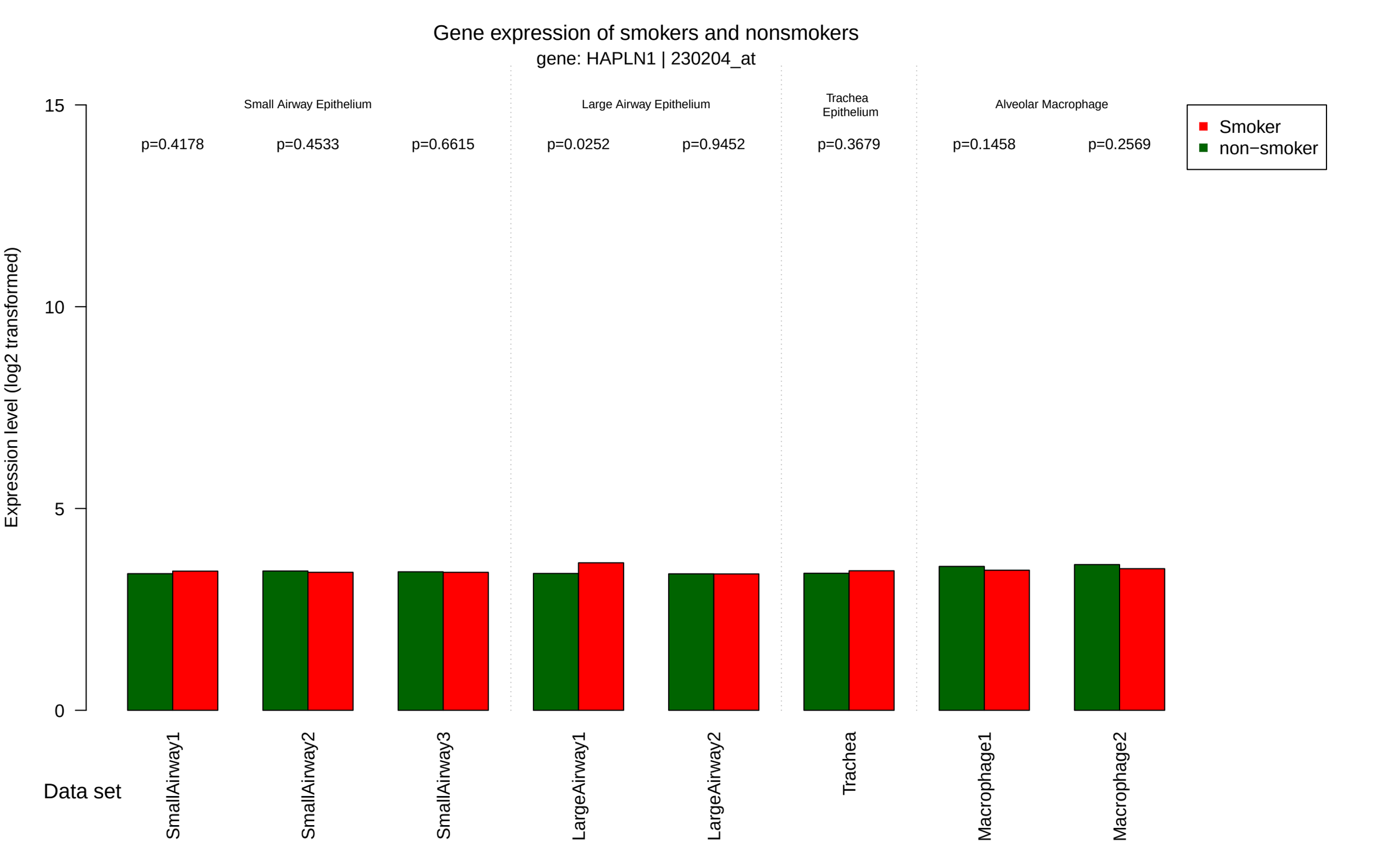
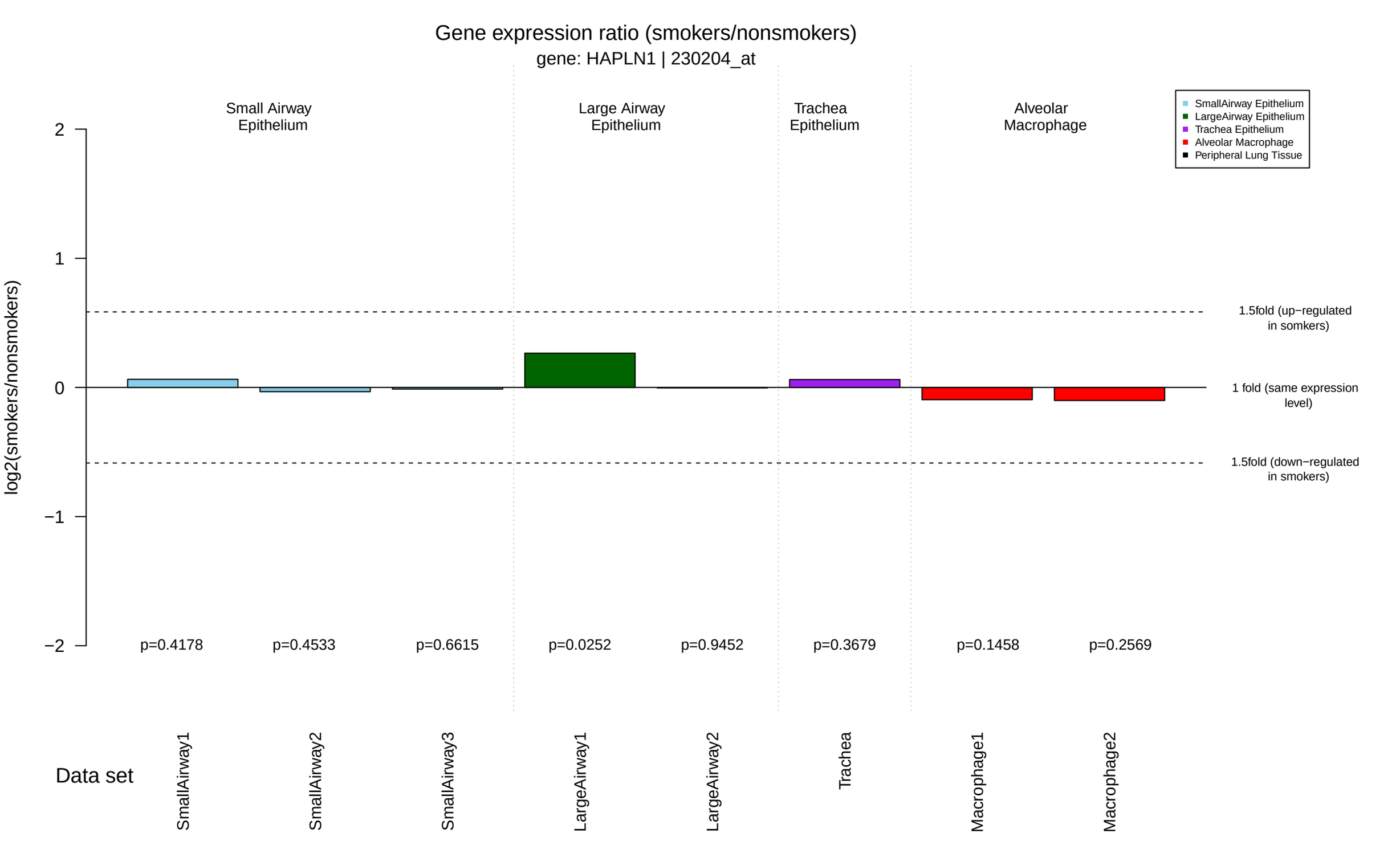
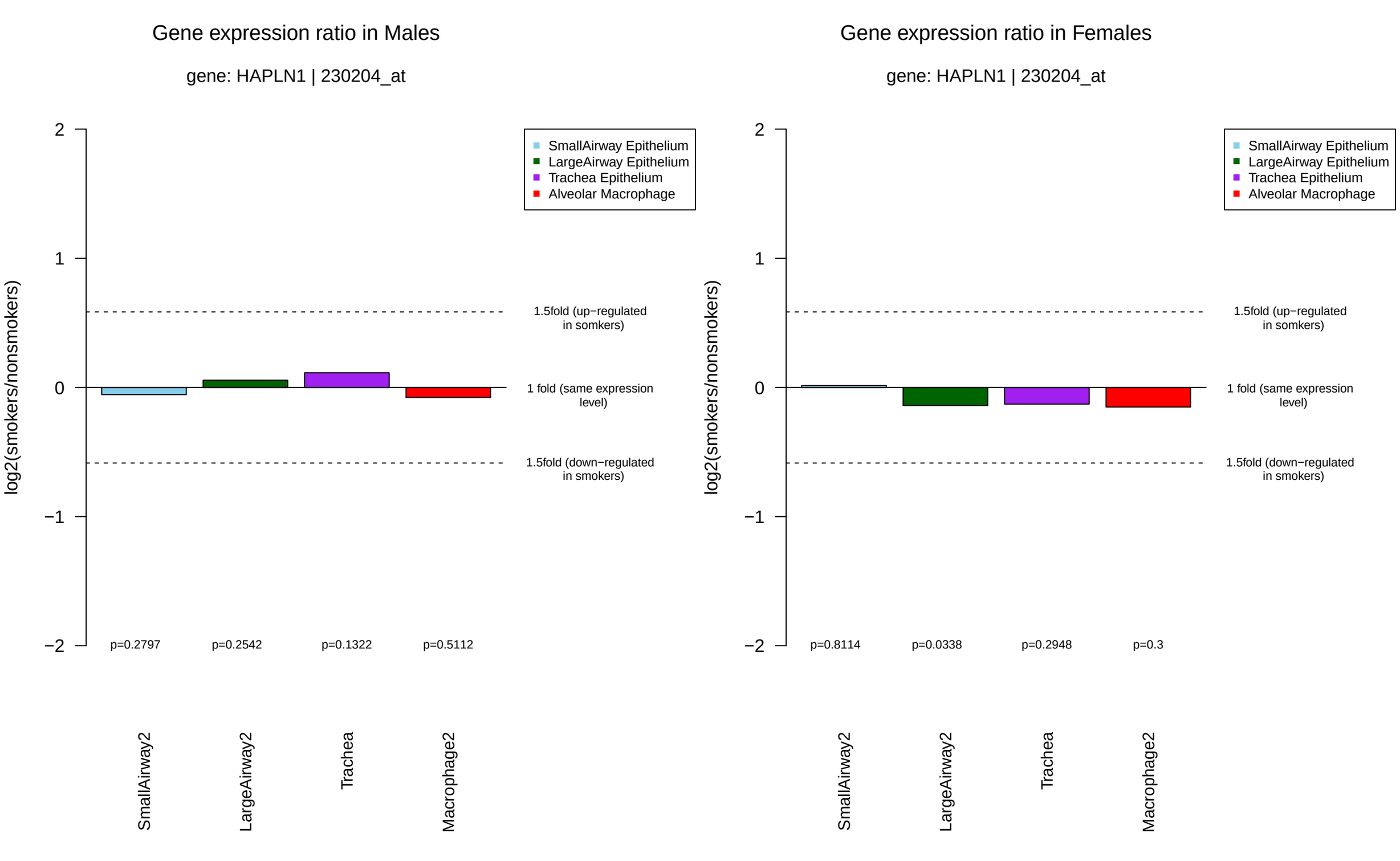
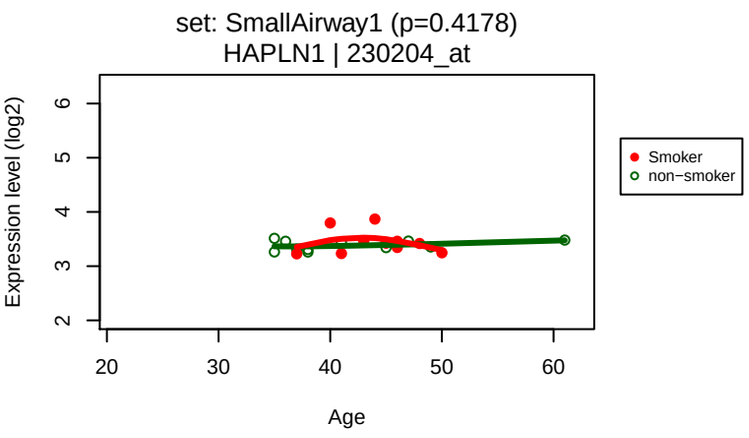
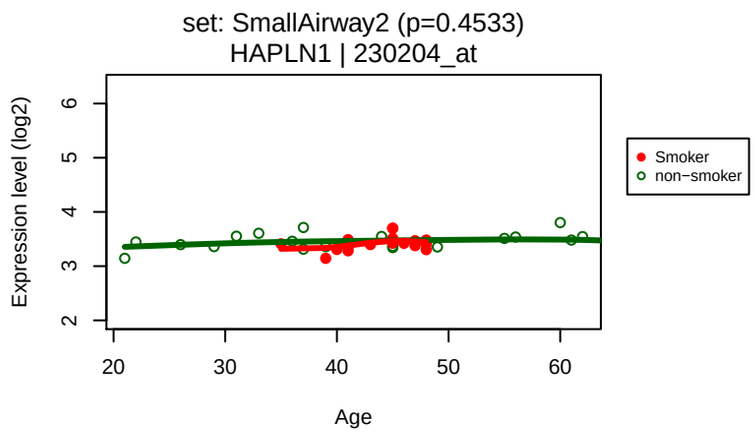
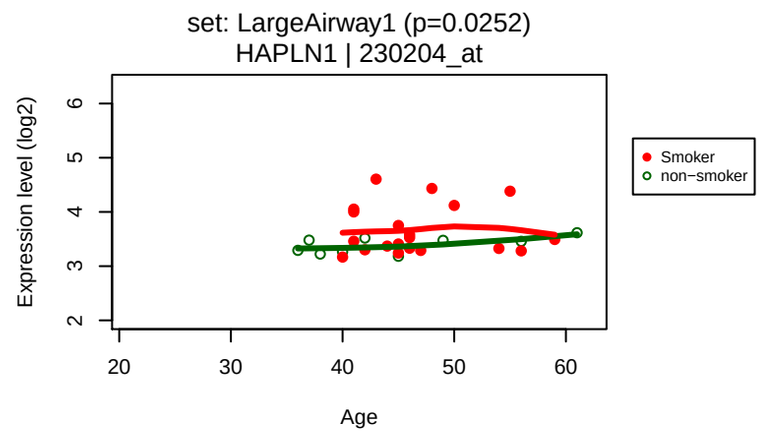

Gene expression of smokers and nonsmokers
gene: HAPLN1 | 230204_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4178
p=0.4533
p=0.6615
p=0.0252
p=0.9452
p=0.3679
p=0.1458
p=0.2569
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

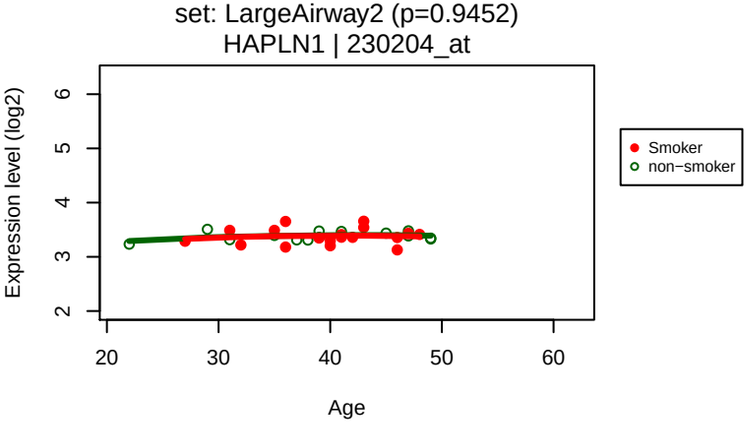

Gene expression ratio (smokers/nonsmokers)
gene: HAPLN1 | 230204_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4178
p=0.4533
p=0.6615
p=0.0252
p=0.9452
p=0.3679
p=0.1458
p=0.2569
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

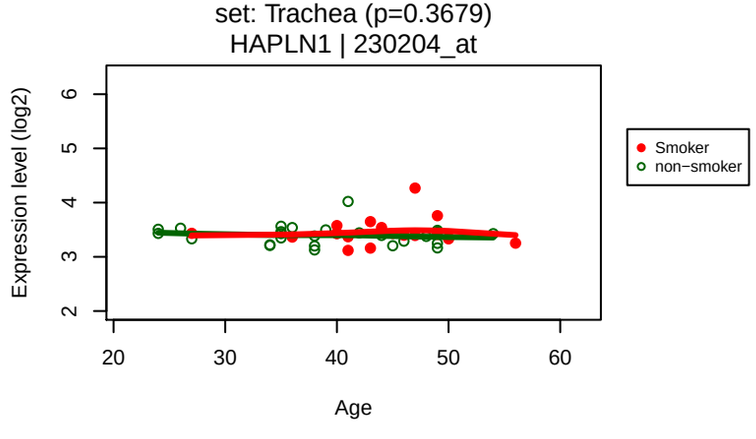

Gene expression ratio in Males
gene: HAPLN1 | 230204_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2797
p=0.2542
p=0.1322
p=0.5112
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HAPLN1 | 230204_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8114
p=0.0338
p=0.2948
p=0.3
SmallAirway2
LargeAirway2
Trachea
Macrophage2

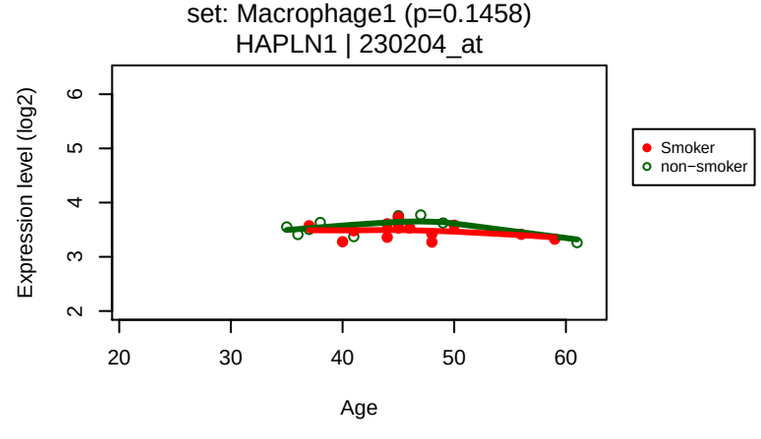

set: SmallAirway1 (p=0.4178)
HAPLN1 | 230204_at
Expression level (log2)
Age
Smoker
non−smoker

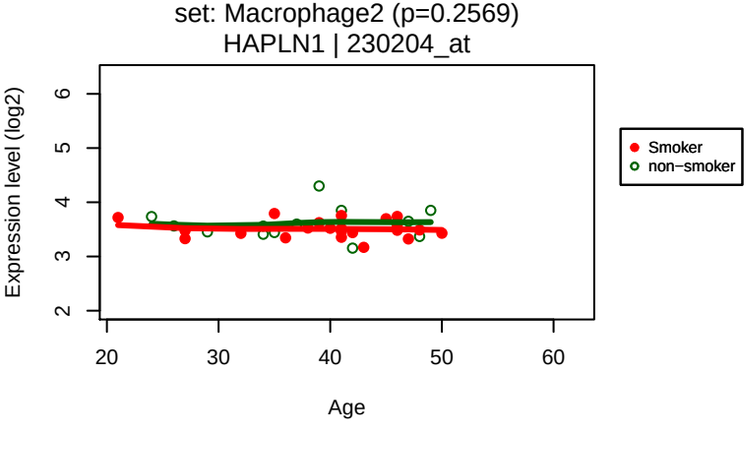

set: SmallAirway2 (p=0.4533)
HAPLN1 | 230204_at
Expression level (log2)
Age
Smoker
non−smoker

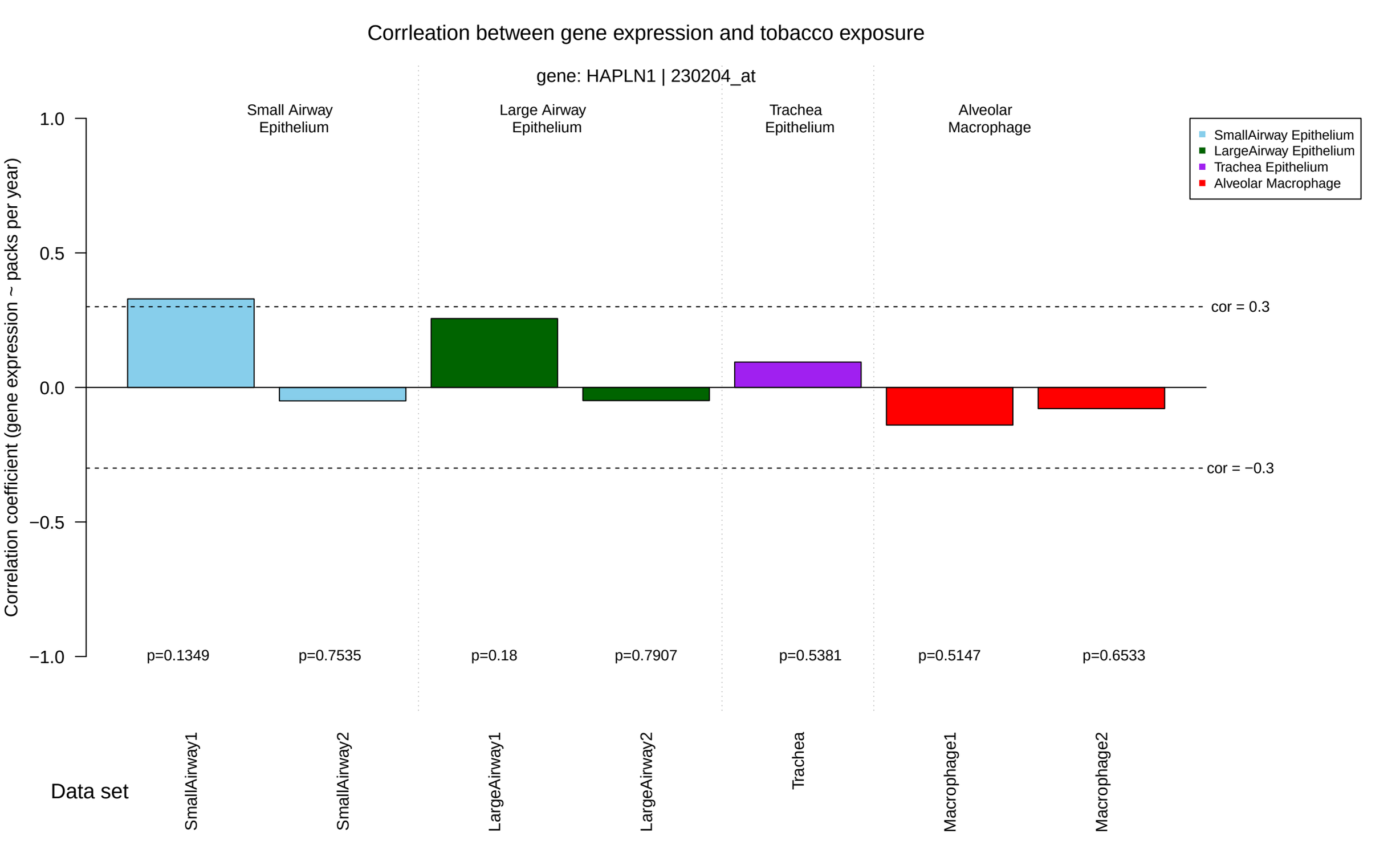

set: LargeAirway1 (p=0.0252)
HAPLN1 | 230204_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9452)
HAPLN1 | 230204_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3679)
HAPLN1 | 230204_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1458)
HAPLN1 | 230204_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2569)
HAPLN1 | 230204_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HAPLN1 | 230204_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1349
p=0.7535
p=0.18
p=0.7907
p=0.5381
p=0.5147
p=0.6533
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2